

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Sampling techniques & measuring abiotic factors - Set 1

Mark scheme

AQA spec 3.7.2.1-3.7.2.2; Research methods; textbook pp. 394-419

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora • 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants); 100 m x 100 m is for trees (impractical, too much work)	2
01.2	• $N = 240$; $N(N-1) = 240 \times 239 = 57\,360$ • $\Sigma n(n-1) = 48 \times 47 + 120 \times 119 + 36 \times 35 + 24 \times 23 + 12 \times 11 = 2256 + 14280 + 1260 + 552 + 132 = 18\,480$ • $D = 57\,360 / 18\,480 = 3.10$ (accept 3.1)	3
01.3	• Lower D means lower biodiversity: the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) • Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread	2
01.4	• A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants • Point quadrat pins are lowered through the vegetation; each touch counts (10 pins = 10% each) without flattening	2
01.5	• Subjective judgement varies between recorders; results are not quantitative so statistical analysis is not possible (but quick for many areas)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
02.1	• Net held with its straight edge on the river bed facing the current (open to the flow) • The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) • Same substrate type, water depth and flow rate at both sites; same net mesh; repeat samples; invertebrates identified with a key and counted	3

Question	Marking guidance	Marks
02.2	<i>Any 3 from:</i> • Difficult to standardise the vigour / area kicked • Buried organisms or those fixed to stones are not dislodged • Some organisms swim and avoid the net • Flow rate varies and affects capture • Fragile organisms destroyed	3
02.3	• A fixed area of bed is enclosed by a frame so the sampled area is standardised; side nets reduce escapes • But only works in fast enough flow and is hard to set on rough substrates / deep water	2
02.4	• Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event • Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
03.1	• Ground surface: pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals; trapped predators eat others; preservative attracts / repels; escapes • Leaf litter: Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light; some die or stop moving; large ones cannot pass the mesh • Branches: beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour; some species not dislodged / fly away; high branches unreachable	6
03.2	• Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught • Brightness / wavelength of the lamp affects attractiveness so catches vary between traps	2
03.3	• The area of ground covered by the suction pipe is known so invertebrates per m² can be calculated • A pooter is only a handling / collection device for individuals already found, not a sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
04.1	• $(60 \times 45) / 9$ • = 300	2
04.2	<i>Any 4 from:</i> • Marked individuals mix freely / randomly with the rest of the population • No births, deaths, immigration or emigration between samples • Marking does not affect survival or behaviour (eg make them more visible to predators) • Marks do not wear off / fall off • All individuals have the same chance of being caught (traps do not attract or repel marked animals)	4
04.3	• Fewer marked beetles would survive to be recaptured • So the proportion marked in the second sample would be too low and the population would be overestimated	2

Question	Marking guidance	Marks
04.4	• Enough time for marked beetles to mix with the population • Not so long that births / deaths / migration or loss of marks change the population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 401-402; AQA spec 3.7.2.1.	
05.1	• Readings drift so they may not be accurate; calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions • pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly • If a meter cannot be adjusted, record the percentage error and correct the results	3
05.2	• Same height above the ground • No obstacles to air flow; impeller type facing into the wind • Axis of rotation horizontal (impeller) or vertical (cup); same time period / averaging	3
05.3	• Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling • Evaporation cools the wet bulb more when humidity is low; the temperature difference is compared with a table to find humidity	2
05.4	<i>Any 2 from:</i> • Test strips with reagent bands compared with a colour chart (semi-quantitative) • Electronic colourimeter: reagent added and colour intensity measured at a specific wavelength • Ion-selective electrode measuring electrode potential (other ions can interfere)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
06.1	• Sieves assembled with the coarsest mesh on top; a dried, crushed soil sample is placed in the top sieve • Stack shaken (about 2 minutes) until no more particles pass; each fraction weighed • Percentages of sand (0.02-2 mm), silt (0.002-0.02 mm) and clay (base pan, below 0.002 mm) calculated; material over 2 mm (stones, twigs) is not soil; plotted on a soil triangle	3
06.2	• Auger collects a core of known cross-sectional area and depth • So samples from different sites are directly comparable (also needed for bulk density = dry mass / volume)	2
06.3	• Minerals such as calcium carbonate break down releasing carbon dioxide • So mass is lost that is not organic matter	2
06.4	• A beam of light passes through the sample in a dark chamber; a light sensor at 90° measures light scattered by suspended particles - more turbid, more scatter • Standardise bulb brightness / colour, distance and container • A calibration curve is produced using samples with known amounts of suspended matter so readings convert to concentrations; Secchi discs / turbidity bottles give only semi-quantitative results	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 410-413, 417; AQA spec 3.7.2.1-3.7.2.2.	
07.1	• Small seeds, genetically uniform, high and rapid germination rate so results are quick and reliable • Wild species have different ranges of tolerance so cross results cannot be applied directly	2
07.2	• Preliminary study with widely spaced pH values to find where germination occurs • Final range from below the lowest pH giving germination to above the highest, with closer intervals	2
07.3	• Temperature (constant, natural germination temperature) • Light, water volume, number of seeds per dish, seed source • Growth medium (filter paper; soil would alter pH); nutrients not needed for germination • Continue until it is clear that ungerminated seeds will not germinate, so that a delay in germination is not mistaken for failure	4
07.4	• Surface water affects fresh mass and drying damages the plants; dry mass requires killing the plants so the same plants cannot be re-measured • Leaf area is a reasonable estimate of biomass that allows repeated measurement of the same plants over time	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 416-418; AQA spec 3.7.2.2.	